

Nikhil Bathula

Senior Software Engineer — Backend, AI & Real-Time Systems

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Hyderabad, India — Open to Remote / Hybrid / Onsite

Professional Summary

Senior Software Engineer with 4+ years of experience designing and building scalable backend platforms, enterprise applications, AI-powered systems, and real-time communication platforms. Experienced in Java Full Stack development, cloud-native architectures, LLM-powered applications, vector databases, and real-time voice AI systems. Strong track record of delivering production-ready solutions with end-to-end ownership, performance optimization, and DevOps automation.

Technical Skills

Programming Languages	Java, JavaScript, TypeScript, Python, Dart, Kotlin, C#
Frontend & Mobile	React, React Native, Flutter, HTML5, CSS3, Tailwind CSS
Backend & APIs	Spring Boot, Node.js, Express, FastAPI, Flask, ASP.NET Core, REST APIs, GraphQL
AI / LLM Technologies	OpenAI GPT-4o, Whisper, Gemini 1.5 Flash, Gemini 2.0 Flash, Claude, Prompt Engineering, Structured Extraction, RAG
Databases	PostgreSQL, MySQL, SQLite, Apache Solr
Vector Databases	Qdrant
Realtime Systems	Twilio Voice SDK, WebSockets, Audio Streaming Pipelines, Voice AI Systems
DevOps & Infrastructure	Docker, Docker Compose, Linux, Nginx, GitHub Actions, CI/CD Pipelines, Hetzner GPU Servers

Professional Experience

D3E Studio India Pvt Ltd

Software Engineer / Senior Software Engineer

Aug 2022 – Present

Hyderabad, India

- Designed and delivered enterprise-grade backend platforms using Java Spring Boot and Node.js supporting web and mobile applications.
- Migrated a PDF generation engine from a Dart-based implementation to a Java backend solution, improving maintainability and scalability.
- Developed and optimized Android and iOS applications using Flutter and React Native.
- Automated deployment pipelines using Docker, Nginx, shell scripting, and CI/CD workflows.
- Designed secure REST APIs, authentication systems, and WebSocket-based real-time communication modules.
- Integrated Twilio Voice SDK for real-time call streaming and voice-based workflows.
- Built AI-powered resume parsing systems and semantic candidate search platforms using OpenAI, vector databases, and Retrieval-Augmented Generation.
- Managed Linux servers and deployment environments for enterprise applications and AI infrastructure.

Key Projects

AI Resume Parser & Semantic Search Platform

- Built an AI-powered pipeline for extracting structured candidate information from PDF and DOCX resumes.
- Implemented multi-step LLM workflows for extracting skills, experience, education, projects, certifications, and metadata.
- Indexed resume embeddings using Qdrant and Pinecone for semantic similarity search.
- Enabled recruiter-focused search, ranking, filtering, and candidate matching.

Real-Time Voice AI Assistant

- Developed a low-latency voice assistant using Twilio Voice and WebSocket streaming.
- Built streaming pipelines connecting Speech-to-Text, LLM inference, and Text-to-Speech systems.
- Integrated OpenAI GPT-4o Realtime API, Whisper, Gemini Live API, and Claude-based workflows.
- Optimized system response times for near real-time conversational experiences.

Low-Code Application Builder Platform

- Engineered reusable code-generation modules supporting Android, iOS, and Web applications.
- Built dynamic widget systems and abstraction layers simplifying multi-platform development.
- Improved developer productivity through automated build and deployment pipelines.

Enterprise HR Management System

- Developed employee lifecycle management, attendance, workflow, and access-control modules.
- Designed scalable backend APIs for enterprise HR operations.
- Improved process automation and centralized employee management.

Enterprise Service & Hostel Management Applications

- Developed scalable service-request tracking and workflow management systems.
- Implemented role-based access control and secure backend services.
- Added offline support and optimized data synchronization mechanisms.

Education

Government Model Residential Polytechnic	2022
Diploma – Electronics & Communication Engineering	

Highlights

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| <ul style="list-style-type: none">• Backend Architecture• AI Integration• Semantic Search• Vector Databases• Real-Time Systems• Twilio Voice | <ul style="list-style-type: none">• OpenAI GPT-4o• Google Gemini• RAG Applications• Performance Optimization• DevOps Automation |
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